Sahaj Bhadja

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Summary

Software engineer with 2+ years of experience building high-impact **Python/C++-oriented distributed systems**, with strength in event-driven workflows, multithreaded processing, quantitative problem solving, and rapid delivery in research-driven engineering environments.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies

PDEU
B.Tech. in Computer Science and Engineering *Nov 2020 – May 2024*
GPA: 3.88/4 Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built and optimized distributed backend services across **FastAPI**, **Azure SQL**, and **PostgreSQL** for a multi-tenant platform serving 500+ users, reducing dashboard latency from 1.2s to 350ms through schema and query optimization.
- Engineered fault-tolerant event-driven background workflows for 5K+ monthly tasks with robust validation and release safeguards, improving throughput 3x while sustaining 99%+ production reliability.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform microservice software with 20+ unit/integration suites using **Jenkins**, **PyTest**, and Mockito, improving repeatability and regression coverage across distributed release cycles.
- Troubleshot event-driven API and ETL workflows using logs, reproducible scenarios, and OOP patterns across **AWS Glue/S3**, improving response times by 30% and pipeline efficiency by 68%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python-based system pipelines for autonomous planning, processing 100K+ trajectory logs and improving long-horizon model performance by 18% through iterative optimization and test-debug cycles.
- Architected spatial-temporal model evaluation flows with attention-based components, improving closed-loop stability by 15% while documenting reproducible diagnostics for team-wide collaboration.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Implemented concurrent processing and robust API workflows for an AI platform using **Flask**, secure ingestion, and **AWS RDS**, handling 10K+ daily requests in production.
- Contributed to definition-through-release delivery with CI/CD-backed validation, reducing turnaround time by 60% while maintaining stable performance under high-load scenarios.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built backend APIs and event-driven synchronization workflows for a booking/payments platform across 50K+ records, covering reservation, pricing, and state-management paths under production constraints.
- Mapped 12+ integration points and collaborated cross-functionally on deployment scenarios, improving reproducibility, stakeholder alignment, and release confidence.

Skills

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL
Systems & Architecture: Event-Driven Architecture, Distributed Systems, Microservices, Concurrency, Multithreaded Programming
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, JSON, OOP, Escalation Triage
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, Data Modeling
Cloud & Infrastructure: AWS (RDS, S3, Glue), Microsoft Azure, Docker, Kubernetes, Jenkins, Git/GitHub
Quantitative & AI: Algorithms, Complexity Analysis, LLM Systems, Prompt Engineering, Reproducible Experimentation
Engineering Practices: Test Driven Development, Unit/Integration Testing, CI/CD, Debugging, Cross-Functional Teamwork, Mentorship Readiness

Projects

CPU-Scheduling Simulator | *C++, React.js, JavaScript*
- Implemented and validated 6 scheduling algorithms across 1K+ simulated process combinations, reducing debugging time by 20% while strengthening operating-system, multithreading, and performance reasoning.

FinNexus | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built an event-driven fintech platform with real-time updates and secure controls, supporting 50K+ users and 1M+ transactions across a distributed deployment stack.

Big-Brain | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Engineered distributed retrieval/indexing pipelines over 50K+ documents with sub-second search performance, improving retrieval latency by 40% through storage and indexing optimization.